

CHE 201 Fall 2019 HW 9

Armaan Mohajir

TOTAL POINTS

96 / 100

QUESTION 1

11 15 / 15

✓ - **0 pts** Correct

- **15 pts** no submission
- **6 pts** v2 calculation incorrect
- **6 pts** P2 calculation incorrect

QUESTION 2

22 15 / 15

✓ - **0 pts** Correct

- **15 pts** Missing answer
- **1 pts** Wrong calculation. $W = -25.8 \text{ kW}$
- **3 pts** $h_1 = 417.46 \text{ kJ/kg}$. $W = -25.8 \text{ kW}$
- **6 pts** $(z_1 - z_2) = -25 \text{ m}$ (negative). $W = -25.8 \text{ kW}$
- **3 pts** $h_3 = 3230.9 \text{ kJ/kg}$. Mass flow rate of steam = 0.0616 kg/s

QUESTION 3

33 20 / 20

✓ - **0 pts** Correct

- **5 pts** power requirement = $-W$
- **20 pts** No submission
- **8 pts** wrong calculation for W
- + **5 pts** Extra points from Question 6

QUESTION 4

44 13 / 15

- **0 pts** Correct

- **15 pts** Missing answer

✓ - **2 pts** Did not define the control volumes

- **2 pts** Did not state the engineering model
- **3 pts** Should have used m_2 to calculate work. $W = 199.9 \text{ hp}$
- **3 pts** $h_1 \sim h_f(180 \text{ }^\circ\text{C}) = 763.22 \text{ kJ/kg}$. $m_2 = 0.371 \text{ kg/s}$

- **1 pts** $m_2 = ?$

- **1 pts** Work is positive

QUESTION 5

55 13 / 15

- **0 pts** Correct

- **15 pts** Missing or wrong answer

✓ - **2 pts** $W_{1-2} = 0.002 \text{ kJ}$

- **7.5 pts** b) $t = 1328.3 \text{ s}$ (22.1 min)

- **2 pts** $W_{2-3} = -0.0522 \text{ kJ}$

- **2 pts** $h_e = 2675.5 \text{ kJ/kg}$

- **3 pts** b) calculation is not clear.

- **13 pts** Incomplete answer

- **2 pts** $u_2 = 417.36 \text{ kJ/kg}$

+ **6 pts** Extra points from Question 6

QUESTION 6

66 20 / 20

✓ - **0 pts** Correct

- **20 pts** Missing answer

+ **20 pts** Extra points for making the board

- **10 pts** Missing or incomplete explanation of how it works and/or missing schematics of important parts (eg: nozzle, fan, switch, heating elements, cord etc).

- **5 pts** Missing source for the drawing

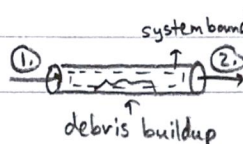
- **2 pts** Missing schematics of the air flow

- **10 pts** Missing drawing

(4.65) Horizontal constant-diameter pipe

- $T_1 = 320\text{ K}$, $P_1 = 900\text{ kPa}$, $V_1 = 30\text{ m/s}$
- $T_2 = 305\text{ K}$

Schematic



Engineering Model

- steady state
- control volume
- one inlet & one outlet
- $\dot{Q}_{cv} = 0$, $\Delta PE = 0$, $\dot{w}_{cv} = 0$
- air is ideal gas

Mass Rate Balance

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

Energy Rate Balance

$$\frac{dE_{cv}}{dt} = 0$$

$$0 = \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) \right] + \dot{Q}_{cv}$$

(a.) Find V_2 (in m/s).

- $h_1 = 320.29\text{ kJ/kg}$
 - $h_2 = 305.22\text{ kJ/kg}$
- Table A-22

$$(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) = 0$$

$$h_1 - h_2 = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$h_1 - h_2 + \frac{V_1^2}{2} = \frac{V_2^2}{2}$$

$$\hookrightarrow V_2 = \sqrt{2(h_1 - h_2) + V_1^2}$$

$$V_2 = \sqrt{2(320.29 - 305.22) \frac{\text{kJ}}{\text{kg}} \cdot \frac{1000\text{ J}}{1\text{ kJ}} + (30)^2}$$

$$V_2 = 176.2\text{ m/s}$$

(b.) Find P_2 (in kPa).

$$\dot{m} = \frac{A V}{v} \quad \& \quad P v = R T$$

$$\frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \rightarrow \frac{A_1 V_1 P_1}{R T_1} = \frac{A_2 V_2 P_2}{R T_2}$$

$$\hookrightarrow P_2 = \left(\frac{T_2}{T_1} \right) \left(\frac{V_1}{V_2} \right) P_1$$

$$P_2 = \left(\frac{305\text{ K}}{320\text{ K}} \right) \left(\frac{30\text{ m/s}}{176.2\text{ m/s}} \right) (900\text{ kPa})$$

$$P_2 = 146.1\text{ kPa}$$

11 15 / 15

✓ - 0 pts Correct

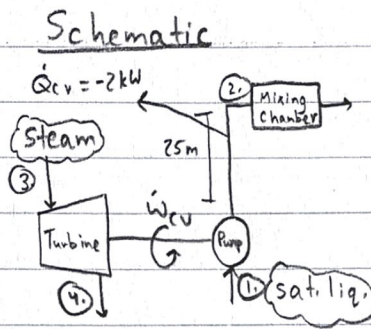
- 15 pts no submission

- 6 pts v2 calculation incorrect

- 6 pts P2 calculation incorrect

(4.70) Turbine-driven pump

- $P_1 = 1 \text{ bar}$, $\dot{m}_1 = 50 \text{ kg/s}$
- $h_2 = 417.69 \text{ kJ/kg}$
- $P_3 = 30 \text{ bar}$, $T_3 = 400^\circ\text{C}$
- $P_4 = 5 \text{ bar}$, $T_4 = 180^\circ\text{C}$



Engineering Model

- control volume
- steady state
- two inlets & two outlets
- $\Delta KE = 0$ everywhere
- $\dot{Q}_{cv} = 0$ & $\Delta PE = 0$ for turbine

(a.) Find $\dot{w}$ (in kW).

$$0 = \dot{m}_1 \left[(h_1 - h_2) + g(z_1 - z_2) \right] + \dot{Q}_{cv} - \dot{w}_{cv}$$

- $h_1 = 417.46 \text{ kJ/kg}$ (Table A-3)

$$0 = 50 \left[(417.46 - 417.69) \frac{\text{kJ}}{\text{kg}} + (9.81 \text{ m/s}^2)(0 - 25) \text{ m} \cdot \frac{1 \text{ N}}{1 \text{ kgm/s}^2} \cdot \frac{1 \text{ kJ}}{1000 \text{ Nm}} \right] - 2 \text{ kW} - \dot{w}_{cv}$$

$$\dot{w}_{cv} = -25.8 \text{ kW}$$

(b.) Find $\dot{m}_3$ (in kg/s).

- $T_3 = 233.90^\circ\text{C}$
 - $h_3 = 3230.9 \text{ kJ/kg}$
 - $T_4 = 151.86^\circ\text{C}$
 - $h_4 = 2812.0 \text{ kJ/kg}$
- Table A-4

$$0 = \dot{m}_3 (h_3 - h_4) - \dot{w}_{cv}$$

$$0 = \dot{m}_3 (3230.9 - 2812.0) \frac{\text{kJ}}{\text{kg}} - 25.8 \text{ kW}$$

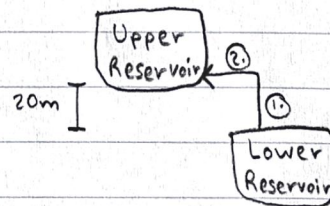
$$25.8 = 418.9 \dot{m}_3 \rightarrow \dot{m}_3 = 0.0615 \text{ kg/s}$$

(4.72) Pumped-hydro energy storage system

Schematic

Engineering Model

- $(A\dot{V})_1 = 150 \text{ m}^3/\text{s}$
- $z_2 = 20 \text{ m}$, $g = 9.81 \text{ m/s}^2$
- $\dot{Q}_{cv} = -0.6 \text{ MW}$



- steady state
- control volume
- one inlet & one outlet
- $\Delta KE = 0$
- No large change in P and T

* Find $\dot{w}$ (in MW)

$h_1 = h_2 \rightarrow$ compressed state

$$0 = \dot{m} \left[(h_1 - h_2) + g(z_1 - z_2) \right] + \dot{Q}_{cv} - \dot{w}_{cv}$$

$$\dot{w}_{cv} = \dot{m} g (z_1 - z_2) + \dot{Q}_{cv}$$

$$\dot{m} = \rho (A\dot{V})_1$$

$$\dot{m} = (1000 \text{ kg/m}^3)(150 \text{ m}^3/\text{s})$$

$$\dot{m} = 150,000 \text{ kg/s}$$

$$\dot{w}_{cv} = (150,000 \text{ kg/s}) \left[(9.81 \text{ m/s}^2)(0 - 20) \text{ m} \right] + (-0.6 \times 10^6 \text{ W})$$

$$\dot{w}_{cv} = -30.03 \text{ MW}$$

2 2 15 / 15

✓ - 0 pts Correct

- 15 pts Missing answer

- 1 pts Wrong calculation. $W = -25.8 \text{ kW}$

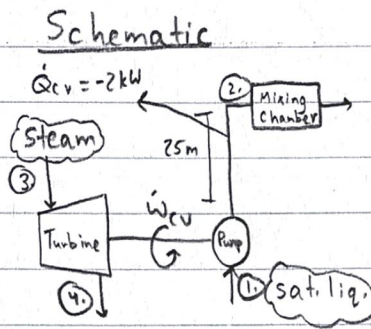
- 3 pts $h_1 = 417.46 \text{ kJ/kg}$. $W = -25.8 \text{ kW}$

- 6 pts $(z_1 - z_2) = -25\text{m}$ (negative). $W = -25.8 \text{ kW}$

- 3 pts $h_3 = 3230.9 \text{ kJ/kg}$. Mass flow rate of steam = 0.0616 kg/s

(4.70) Turbine-driven pump

- $P_1 = 1 \text{ bar}$, $\dot{m}_1 = 50 \text{ kg/s}$
- $h_2 = 417.69 \text{ kJ/kg}$
- $P_3 = 30 \text{ bar}$, $T_3 = 400^\circ\text{C}$
- $P_4 = 5 \text{ bar}$, $T_4 = 180^\circ\text{C}$



Engineering Model

- control volume
- steady state
- two inlets & two outlets
- $\Delta KE = 0$ everywhere
- $\dot{Q}_{cv} = 0$ & $\Delta PE = 0$ for turbine

(a.) Find $\dot{w}$ (in kW).

$$0 = \dot{m}_1 \left[(h_1 - h_2) + g(z_1 - z_2) \right] + \dot{Q}_{cv} - \dot{w}_{cv}$$

- $h_1 = 417.46 \text{ kJ/kg}$ (Table A-3)

$$0 = 50 \left[(417.46 - 417.69) \frac{\text{kJ}}{\text{kg}} + (9.81 \text{ m/s}^2)(0 - 25) \text{ m} \cdot \frac{1 \text{ N}}{1 \text{ kgm/s}^2} \cdot \frac{1 \text{ kJ}}{1000 \text{ Nm}} \right] - 2 \text{ kW} - \dot{w}_{cv}$$

$$\dot{w}_{cv} = -25.8 \text{ kW}$$

(b.) Find $\dot{m}_3$ (in kg/s).

- $T_3 = 233.90^\circ\text{C}$
 - $h_3 = 3230.9 \text{ kJ/kg}$
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 - $h_4 = 2812.0 \text{ kJ/kg}$
- Table A-4

$$0 = \dot{m}_3 (h_3 - h_4) - \dot{w}_{cv}$$

$$0 = \dot{m}_3 (3230.9 - 2812.0) \frac{\text{kJ}}{\text{kg}} - 25.8 \text{ kW}$$

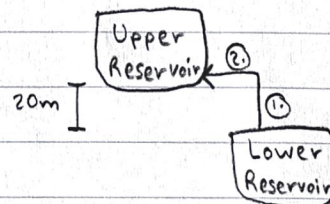
$$25.8 = 418.9 \dot{m}_3 \rightarrow \dot{m}_3 = 0.0615 \text{ kg/s}$$

(4.72) Pumped-hydro energy storage system

Schematic

Engineering Model

- $(A\dot{V})_1 = 150 \text{ m}^3/\text{s}$
- $z_2 = 20 \text{ m}$, $g = 9.81 \text{ m/s}^2$
- $\dot{Q}_{cv} = -0.6 \text{ MW}$



- steady state
- control volume
- one inlet & one outlet
- $\Delta KE = 0$
- No large change in P and T

* Find $\dot{w}$ (in MW)

$h_1 = h_2 \rightarrow$ compressed state

$$0 = \dot{m} \left[(h_1 - h_2) + g(z_1 - z_2) \right] + \dot{Q}_{cv} - \dot{w}_{cv}$$

$$\dot{w}_{cv} = \dot{m} g(z_1 - z_2) + \dot{Q}_{cv}$$

$$\dot{m} = \rho(A\dot{V})$$

$$\dot{m} = (1000 \text{ kg/m}^3)(150 \text{ m}^3/\text{s})$$

$$\dot{m} = 150,000 \text{ kg/s}$$

$$\dot{w}_{cv} = (150,000 \text{ kg/s}) \left[(9.81 \text{ m/s}^2)(0 - 20) \text{ m} \right] + (-0.6 \times 10^6 \text{ W})$$

$$\dot{w}_{cv} = 30.03 \text{ MW}$$

33 20 / 20

✓ - 0 pts Correct

- 5 pts power requirement = -W

- 20 pts No submission

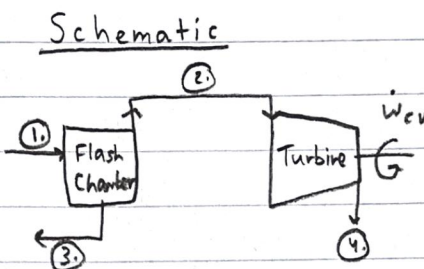
- 8 pts wrong calculation for W

+ 5 pts Extra points from Question 6

(4.75) Flash chamber to turbine

- $P_1 = 15 \text{ bar}$, $T_1 = 180^\circ\text{C}$, $\dot{m}_1 = 5 \text{ kg/s}$
- $P_2 = 4 \text{ bar} \rightarrow \text{Saturated vapor}$
- $P_3 = 4 \text{ bar} \rightarrow \text{Saturated liquid}$
- $P_4 = 0.08 \text{ bar}$, $x_4 = 90\%$

* Find $\dot{w}$ (in hp) of the turbine.



Engineering Model

- steady state
- control volume
- two inlets & two outlets
- $\dot{Q}_{cv} = 0$, $\Delta KE = 0$, & $\Delta PE = 0$

Mass Rate Balance

$$\dot{m}_1 - \dot{m}_2 - \dot{m}_3 = 0$$

Energy Rate Balance

$$0 = \dot{m}_1 \left(h_1 + \frac{V_1^2}{2} + gz_1 \right) - \dot{m}_2 \left(h_2 + \frac{V_2^2}{2} + gz_2 \right) - \dot{m}_3 \left(h_3 + \frac{V_3^2}{2} + gz_3 \right)$$

$$0 = \dot{m}_1 h_1 - \dot{m}_2 h_2 - \dot{m}_3 h_3 \rightarrow \dot{m}_1 h_1 = \dot{m}_2 h_2 + \dot{m}_3 h_3$$

$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3$$

$$\dot{m}_3 = \dot{m}_1 - \dot{m}_2$$

$$\dot{m}_1 h_1 = \dot{m}_2 h_2 + (\dot{m}_1 - \dot{m}_2) h_3$$

$$\dot{m}_1 h_1 = \dot{m}_2 h_2 + \dot{m}_1 h_3 - \dot{m}_2 h_3$$

$$\dot{m}_1 h_1 - \dot{m}_1 h_3 = \dot{m}_2 h_2 - \dot{m}_2 h_3$$

$$\dot{m}_2 = \frac{\dot{m}_1 (h_1 - h_3)}{(h_2 - h_3)}$$

$$\cdot h_1 = 763.22 \text{ kJ/kg} = h_f \text{ (Table A-2)}$$

$$\cdot h_3 = 604.74 \text{ kJ/kg} = h_f \text{ (Table A-3)}$$

$$\cdot h_2 = 2738.6 \text{ kJ/kg} = h_g \text{ (Table A-3)}$$

$$\dot{m}_2 = \frac{(5 \text{ kg/s}) (763.22 - 604.74) \text{ kJ/kg}}{(2738.6 - 604.74) \text{ kJ/kg}}$$

$$\dot{m}_2 = 0.371 \text{ kg/s}$$

$$\left. \begin{array}{l} h_f = 173.88 \text{ kJ/kg} \\ h_g = 2577.0 \text{ kJ/kg} \end{array} \right\} @ P_4 = 0.08 \text{ bar} \text{ Table A-3}$$

$$\cdot h_4 = (1-x) h_f + x h_g$$

$$h_4 = (1-0.90) (173.88 \text{ kJ/kg}) + (0.90) (2577.0 \text{ kJ/kg})$$

$$h_4 = 2336.688 \text{ kJ/kg}$$

$$0 = \dot{m}_2 (h_2 - h_4) - \dot{w}_{cv} \rightarrow \text{Turbine}$$

$$\dot{w}_{cv} = \dot{m}_2 (h_2 - h_4)$$

$$\dot{w}_{cv} = (0.371 \text{ kg/s}) (2738.6 - 2336.688) \frac{\text{kJ}}{\text{kg}} \cdot \frac{1 \text{ hp}}{0.7457 \text{ kJ/s}}$$

$$\dot{w}_{cv} = 199.96 \text{ hp}$$

4 4 13 / 15

- 0 pts Correct
- 15 pts Missing answer
- ✓ - 2 pts Did not define the control volumes
 - 2 pts Did not state the engineering model
 - 3 pts Should have used m_2 to calculate work. $W = 199.9$ hp
 - 3 pts $h_1 \sim h_f(180 \text{ oC}) = 763.22$ kJ/kg. $m_2 = 0.371$ kg/s
 - 1 pts $m_2 = ?$
 - 1 pts Work is positive

(4.87) Open cooking pot

- $P_1 = 1 \text{ bar}$, $T_1 = 20^\circ\text{C}$, $V_1 = 0.5 \text{ L}$
- $\dot{Q}_{cv} = 0.85 \text{ kW}$, $P_2 = P_1$

(a) Find the time required for the onset of evaporation (in s).

• $u_1 = 83.8 \text{ kJ/kg}$
• $v_1 = 0.0010 \text{ m}^3/\text{kg}$ } Table A-2 $m_1 = \frac{0.5 \times 10^{-3} \text{ m}^3}{0.0010 \text{ m}^3/\text{kg}} \rightarrow m_1 = 0.5 \text{ kg}$

* $v = \frac{V}{m} \rightarrow m = \frac{V}{v}$

• $u_2 = 417.3 \text{ kJ/kg}$
• $v_2 = 0.0010 \text{ m}^3/\text{kg}$ } Table A-2 $W = m \int_{1-2}^2 P dv \rightarrow W = mP(v_2 - v_1)$
 $W = (0.5 \text{ kg})(100 \text{ kPa})(0.0010 - 0.0010) \text{ m}^3/\text{kg}$
 $W = 0$

* $\Delta U = Q - W$ $\dot{Q} \Delta t = m(u_2 - u_1) + W$ $\Delta t = \frac{(0.5 \text{ kg})(417.3 - 83.8) \text{ kJ/kg} + 0}{(0.85 \text{ kW})}$
 $Q = \Delta U + W$ $\Delta t = \frac{m(u_2 - u_1) + W}{\dot{Q}}$
 $\Delta t = 196.175$

(b) Find the time required for all of the water to evaporate once evaporation starts (in s).

Mass Rate Balance

$$\dot{m}_3 - \dot{m}_2 = 0$$

$$W_{2-3} = m \int_2^3 P dv \rightarrow W_{2-3} = mP(v_3 - v_2)$$

$$W_{2-3} = (0.5 \text{ kg})(100 \text{ kPa})(0 - 0.0010) \text{ m}^3/\text{kg}$$

$$W_{2-3} = -0.05 \text{ kJ}$$

$$\Delta t = - \frac{(\dot{m}_3 - \dot{m}_2) h_{in} + (\dot{m}_3 u_3 - \dot{m}_2 u_2) + W}{\dot{Q}}$$

• $h_{in} = 2675.1 \text{ kJ/kg}$
• $\dot{m}_3 = 0$

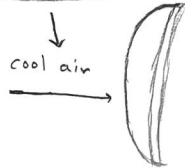
$$\Delta t = - \frac{(0 - 0.5) \text{ kg/s} (2675.1 \text{ kJ/kg}) + (0 - (0.5 \text{ kg/s})(417.3 \text{ kJ/kg})) - 0.05 \text{ kJ}}{(0.85 \text{ kW})}$$

$$\Delta t = 1328.058 \text{ s}$$

5 5 13 / 15

- 0 pts Correct
- 15 pts Missing or wrong answer
- ✓ - 2 pts **W 1-2 = 0.002 kJ**
- 7.5 pts b) $t = 1328.3 \text{ s}$ (22.1 min)
- 2 pts $W_{2-3} = -0.0522 \text{ kJ}$
- 2 pts $h_e = 2675.5 \text{ kJ/kg}$
- 3 pts b) calculation is not clear.
- 13 pts Incomplete answer
- 2 pts $u_2 = 417.36 \text{ kJ/kg}$
- + 6 pts Extra points from Question 6

Cool air passes through the entry point indicated below so that it can reach the motor.



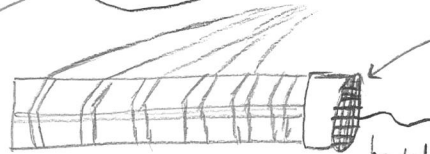
Electricity is used as the power source and the fan blades bring in cool air; the fan is powered by the motor.

fan blades



motor

The motor directly powers the fan blades, which in turn passes the room temperature air towards the heating coil.



wire coils

The air is heated via a metal coil; this heating coil takes the produced electric energy and converts it into convective heat energy.

heated air

Mass Rate Balance

$$\dot{m}_i - \dot{m}_e = 0$$

Energy Rate Balance

$$0 = \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right) + \dot{Q} - \dot{W}$$

Armaan Mohajir
11/8/19

CHE 201 HW 9 (UAD)

Switch

The switch is used to turn the hair dryer on & off.

cord

Electricity passes through the cord above.

* The motor and the wire coils are the two main components of a handheld hair dryer. The heat transfer process present in this device is rather simple, as electric energy is transformed into convective heat (convection is defined as the heat transfer present in a system due to the movement of fluid particles).

6 6 20 / 20

✓ - 0 pts Correct

- 20 pts Missing answer

+ 20 pts Extra points for making the board

- 10 pts Missing or incomplete explanation of how it works and/or missing schematics of important parts (eg: nozzle, fan, switch, heating elements, cord etc).

- 5 pts Missing source for the drawing

- 2 pts Missing schematics of the air flow

- 10 pts Missing drawing